Locally Linear Unbiased Randomization Network for Cross-Scene Hyperspectral Image Classification

Hanqing Zhao, Jiawei Zhang, Lianlei Lin, Junkai Wang, Sheng Gao, and Zongwei Zhang

***Abstract*— For hyperspectral cross-domain recognition applications, the unseen target domain (TD) is inevitable, and the model can only be trained on the source domain (SD) but directly applied to unknown domains. A major challenge of this domain generalization (DG) problem comes from the domain shift caused by differences in environments, devices, etc. One feasible strategy is performing domain expansion with latent variables and learning domain-invariant representation. Inspired by this framework, the study proposes a generation network for extension, which consists of a symmetric encoder-decoder to implicitly build local joint feature under style randomization. Moreover, supervised contrastive learning is employed to avoid duplicate augmentation. Besides, considering the trade-off between domain-specific and domain-invariant, an adversarial penalty term is formed by inter-class and intra-class contrastive regularization in the discriminator. Multiple evaluations on three public HSI datasets indicate that the proposed method outperforms state-of-the-art (SOTA) approaches. The codes is available from the website: https://github.com/HUOWUMO/IEEE_HSIC_LLURnet.**

***Index Terms*— Contrastive learning, domain extension, domain generalization (DG), hyperspectral image (HSI) classification.**

ABBREVIATIONS DESCRIPTION

Abbreviation	Description
SD	Source domain.
TD	Target domain.
ED	Extension domain.
IS	Intermediate samples.
DA	Domain adaptation.
DG	Domain generalization.
G	Generator.
D	Discriminator.

I. INTRODUCTION

HYPERSPECTRAL image (HSI) classification (HSIC) has obtained excellent performance in independent and identically distributed (iid) scene using convolutional neural networks (CNNs) [1], [2], [3]. However, practical applications of iid are rare because the region to be evaluated is often unknown. These applications specifically need to learn from labeled data SD, like the HSI of several cities, to classify unlabeled data with the same label space TD [4], such as the HSI of other cities, and ultimately develop a model with shared knowledge transfer capabilities. The main challenge is that the HSI is unavoidably influenced by a number of factors, including sensor nonlinearities, seasonal, or meteorological conditions [5], [6], resulting in variances in spectrum reflection between SD and TD.

To reduce the distribution difference between SD and TD in the cross-scene classification challenge, the DA method is primarily developed [7], [8], [9]. In the DA situation, the samples with labeling are referred to as the SD, while the data to be estimated with few or no labels is called the TD. The ultimate objective is to transfer the information from an SD to further improve TD categorization tasks. DA techniques that align the marginal distribution, conditional distribution, or both between two domains are essentially divided into the following three categories: instance-based, classifier-based, and feature-based approaches [10]. However, whenever a new TD is obtained, the DA algorithm has to restart the training process. On the one hand, for HSI with hundreds of spectral channels, the cost of data storage is huge. On the other hand, the extraordinary demand for computing resources will shut out applications with real-time requirements or limited computing resources.

In task setting, DG is more general and demanding than DA, because training samples only consist of labeled SD. DG aims at discovering a model from one or more distinct but related domains that transfer effectively to TD [11]. In the field of generalized image processing, DG has been continuously studied in computer vision (CV) [12], [13], [14], due to the differences in imaging and information distribution between HSI and RGB images (for example, the category information of HSI is mainly concentrated in the spectral dimension rather than spatial one [15]). Therefore, the advanced classification algorithms in CV are not suitable enough for HSIC, which can be proved in the comparative experiments from [16], [17], and [18].

Only a tiny number of works now concentrate on the DG of HSIC. The single SD expansion network (SDEnet [17]) demonstrates that DG is more practical than typical DA, which proposes a semantic encoder integrating spatial-spectral information and a morph encoder for guaranteeing both the efficacy and dependability of generated data. The language-aware DG network (LDGnet [18]) proposes a language-guided framework, which constrains the domain-invariant representation through image and text encoder.

Despite the fact that these algorithms are capable of generalization. The common feature statistical structure, which

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is influenced by DG algorithms in CV [19], [20], tends to represent marginal distributions and excessively compress HSI inline information, such as SDEnet, which includes distinct semantic and morph encoder to mimic RGB image processing. Another problem is that popular convolution layers, such as the residual block, imply a strong inductive bias of image styles, which is sensitive to domain changes [21].

In this study, to address the feature distribution modeling problem, a generative model based on 3-D convolution is developed, where the latent variables representing the spatial-spectral joint probability distribution are examined to reduce the domain gap, and new samples generated by SD are referred to as the extended domain (ED). Furthermore, in light of the CNN inductive bias, a style random layer is proposed to implement feature shuffling and fusion. The obtained unbiased style cubes are decoded by the inverse process of the encoder.

The generator's backbone contains the above encoder-randomization-decoder is now complete, but two issues remain. The first is that the information amount for categorization in each spectrum band varies between datasets recorded by different HSI cameras. Yet, because the receptive field of the convolution step is always fixed, the network may be sensitive to certain spectral bands. To desensitize the generator to spectral information density, aggregated resampling is performed on the locally linear embedding. The other issue is that, even though randomization smooths the latent space preference, the generator may still be optimized to obtain a meaningless solution, resulting in a significant overlap between the generated ED and the SD, so data augmentation does not bring more domain invariance information to the discriminator. To ensure nonredundant generation, an intraclass comparison penalty item is used.

To ensure the class separability of feature extraction in the discriminator, inter-class supervised contrastive learning is developed to cater to Cluster Assumptions and Low-density Separation Assumptions [22], but an unrestricted comparison between classes may bring the risk of excessive aggregation or separation of SD and ED, so intraclass comparison is introduced again and constitutes a new confrontation training.

The following is a summary of the main contributions of the proposed work (LLURnet).

1) A patch-wise style unbiased generator is proposed, which is more suitable for HSI, its encoder-randomize-decoder architecture provides generation efficacy and reliability even with small percentages of training samples.
2) A double-adversarial regularization training for the generator and discriminator is suggested, which accomplishes nonredundant domain expansion and alleviates the contradiction between aggregation and separation in feature space.
3) The whole framework achieves superior generalization performance with fairly low computational complexity.

The remainder of the article is structured as follows. Section II introduces key DG and contrastive learning theories. Section III describes the LLURnet in in-depth detail. Section IV describes the comprehensive experiments and analyses. Section V concludes with conclusions. Supplementally, for the convenience of reading, the important abbreviations are summarized in Nomenclature.

II. Related Work

A. Domain Generalization

The DG methods is presented to overcome the domain shift and also the lack of target data [23]. The main purpose is to learn a model from information collected from a single or several related yet separate SD so that the model may generalize effectively to any out-of-domain (OOD) TD. Existing DG approaches are classified into three categories: data manipulation, representation learning, and learning strategies. Data manipulation is the technique of improving training data by enriching and modifying samples, and divided into two parts: data augmentation [24], [25], and data production [12], [26]. Learning domain-invariant representation is referred to as representation learning, because it allows the model to be properly fit to diverse domains. The kernel method [27], explicit feature alignment [28], [29], domain adversarial training [30], [31], and invariant risk minimization (IRM) [32] are the four fundamental components of domain invariant feature learning. The goal of domain-invariant feature learning is satisfied by feature decoupling. Learning strategy refers to the incorporation of mature learning models into multidomain training to improve model generalization, which primarily includes ensemble learning [33] and meta-learning [34].

B. Contrastive Learning

Contrastive learning focuses on learning the common features between instances of the same kind and distinguishing the differences between different instances. Compared with generative learning, contrastive learning does not need to pay attention to the cumbersome details, but only needs to learn distinguish data in the feature space of the abstract semantic level, so the model and its optimization become simpler and the generalization ability is stronger. For any sample x, contrastive methods aim to learn an feature extractor F such that:

In contrastive learning, the emphasis is on identifying the similarities among instances of the same category as well as the contrasts among them. Contrastive learning differs from generative learning in that it does not require concentration on tedious details, instead, it simply requires learning to discriminate information within the feature space of the generic semantic degree. As a result, the model and optimization are simpler, and the generalization capacity is substantially more robust. Contrastive approaches seek to develop a feature extractor F in the following equation, for any sample x:

$$S\left(F(\mathbf{x}), F\left(\mathbf{x}^{+}\right)\right) \gg S\left(F(\mathbf{x}), F\left(\mathbf{x}^{-}\right)\right). \tag{1}$$

$\mathbf{x}^{+}$ is set to a positive sample that is similar to $\mathbf{x}$, $\mathbf{x}^{-}$ is set to a negative sample that is dissimilar to $\mathbf{x}$, S is a metric that calculates the correlation between two features, and $\mathbf{x}$ is generally called as an "anchor." The InfoNCE loss [35] is

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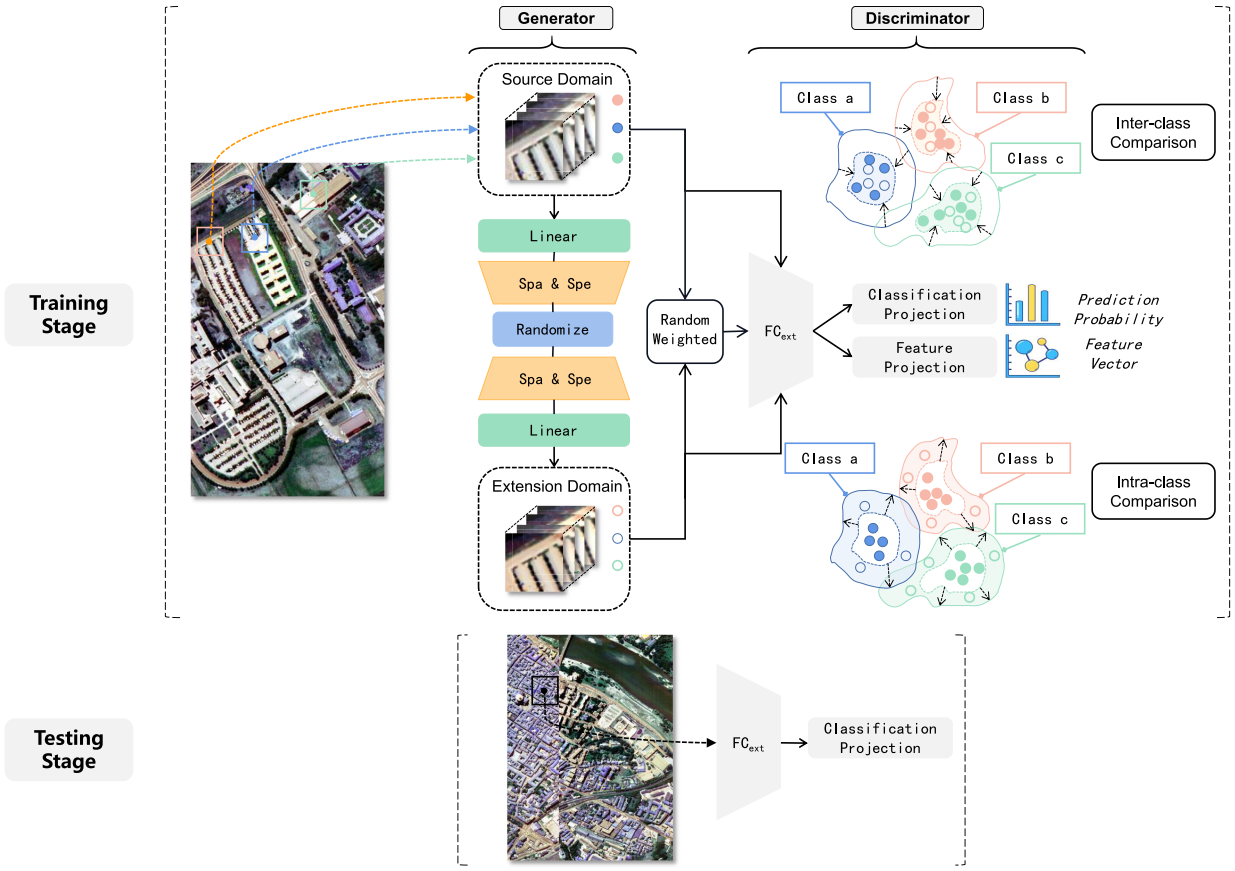


Fig. 1. Flowchart of the LLURnet for training, including generator and discriminator. For testing, only part of the discriminator for classification projection is kept. The embedding features of SD output by the feature projection are shown as solid circles, while the embedding features of ED and random weighted results are shown as hollow circles.

commonly used to optimize for this attribute by effectively separating positive and negative samples.

III. Proposed DG Network

As shown in Fig. 1, The proposed network consists of a patch-wise generator for data augmentation and a discriminator for learning domain-invariant features. Throughout the training process, the two are cross-optimized, and only the discriminator's category prediction structure is employed in testing. The generator and accompanying feature space regularization constitute the network's core. The discriminator is simply a basic convolution classification network with poor generalization capabilities without the generator. (see the $\mathrm{FC_{ext}}$ in ablation experiment of Table IV).

A. Generator

As shown in Fig. 2, The generator consists of a symmetric encoding-randomization-decoding structure.

1) Encoder: Including linear resampling and 3-D convolution-based style extraction, the HSI patch (size of $13 \times 13 \times N_{\text{band}}$, the label of the center pixel as the label of patch) first performs spectral channel scaling (resampling) through 2-D convolution with 1×1 kernel, obtain a 3-D matrix of $13 \times 13 \times \dim_1$, aim to maintain the linear manifold structure of the input, while decoupling the model from the spectral information density, and then use $3 \times 3 \times 3$ kernel to extract the space-spectrum local features, which are considered as a biased style cube set ($\mathbf{z}$). The size of each cube is $11 \times 11 \times (\dim_1 - 2)$, and the number is $\dim_2$.

2) Randomization: The randomization is to eliminate style bias for cubes. The first process is caculating its own mean (μ) and variance (σ) in (2) and (3), where $[H, W, D] = [11, 11, \dim_1 - 2]$. Then assign a new mean ($\hat{\mu}$) and variance ($\hat{\sigma}$) to each cube in (4) and (5), where $\mathbf{z}'$ is the cube set after randomly disruption and α is the adaptive weight. Finally, the unbiased style cube set is calculated by (6)

$$\mu(\mathbf{z}) = \frac{1}{HWD}\sum_{h=1}^{H}\sum_{w=1}^{W}\sum_{d=1}^{D}\mathbf{z} \tag{2}$$

$$\sigma(\mathbf{z}) = \sqrt{\frac{1}{HWD}\sum_{h=1}^{H}\sum_{w=1}^{W}\sum_{d=1}^{D}(\mathbf{z}-\mu(\mathbf{z}))^2} \tag{3}$$

$$\hat{\mu} = \alpha\mu(\mathbf{z}) + (1-\alpha)\mu(\mathbf{z}') \tag{4}$$

$$\hat{\sigma} = \alpha\sigma(\mathbf{z}) + (1-\alpha)\sigma(\mathbf{z}') \tag{5}$$

$$\mathrm{Ran}(\mathbf{z}, \mathbf{z}') = \hat{\sigma}\frac{\mathbf{z}-\mu(\mathbf{z})}{\sigma(\mathbf{z})} + \hat{\mu}. \tag{6}$$

3) Decoder: The decoding procedure is designed as the inverse of encoding, with the unbiased style cube being upsampled by inverse 3DCNN (DeConv3D) and inverse 2DCNN (DeConv2D) to restore the input patch size. It should be noted

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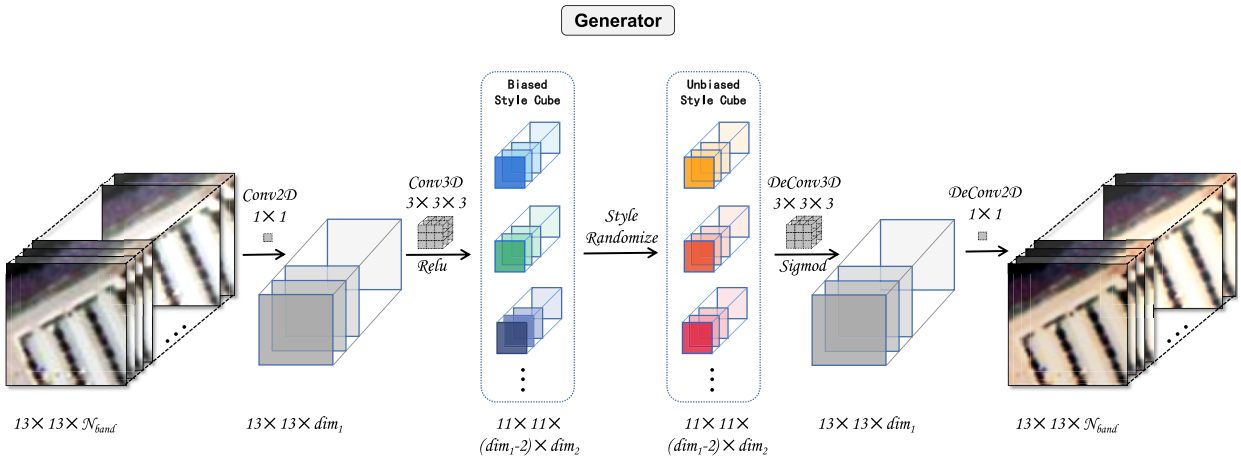


Fig. 2. Flowchart of the generator.

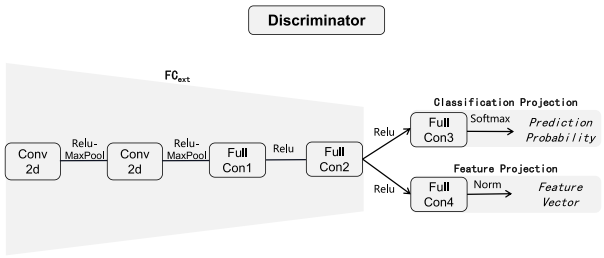


Fig. 3. Flowchart of the discriminator.

that the decoder does not have to be structurally antagonistic to the encoder for the purposes of reconstruction only. The symmetrical structure is intended to ensure that the encoding and decoding have consistent expressive capacities without damaging the feature space's potential correlation.

4) Regularization: In order to optimize G, constraining ED in the correct label space is essential. More importantly, intra-class comparison learning in (7) is developed to enhance the effectiveness to generate ED. The n_c is the number of samples of the cth class. The samples of the cth class in SD are regarded as positive ones, which the number is $|P(i)|$, and the samples of the cth class in ED or IS are regarded negative ones. $P_c(i)$ and $N_c(i)$ are the positive and negative sets of the cth class, respectively. In Each iteration, negative sample sets are formed by randomly selecting from ED or IS. Additionally, the $P_c(i)$ and $N_c(i)$ supervision labels for the cth class are positioned at 0 and 1, respectively

$$\mathcal{L}_2 = -\sum_{c}\sum_{i=0}^{n_c}\frac{1}{|P_c(i)|}\sum_{p\in P_c(i)}\log\frac{\exp\left(S\left(\mathbf{z}_i,\mathbf{z}_p^+\right)/\tau\right)}{\sum_{a\in N_c(i)}\exp\left(S\left(\mathbf{z}_i,\mathbf{z}_n^-\right)/\tau\right)}. \tag{7}$$

B. Discriminator

1) Network: As described in Fig. 3, A classification projection C, a feature projection F, and a feature extractor FC_{ext} constitute the discriminator D. In (8), where y_i is the one-hot encoding of x_i, c is the index of class, and p_i is the predicted probability calculated by C. In (9) and (10), $\mathbf{Y}$ and N are the SD and ED labels and their number.

Furthermore, the randomly weighted from SD and ED outcomes in (11) serve as ISs, as a transition between SD and ED that additionally be accurately predicted to lessen the demand on D to acquire ED domain features [17]. They have the same label space as SD, the label $\mathbf{Y}$ of SD is utilized to calculate the cross entropy loss of ED in (10) and IS in (12)

$$\mathcal{L}_{\mathcal{CE}}(\mathbf{p}_i,\mathbf{y}_i) = -\sum_{c} y_i^c \log p_i^c \tag{8}$$

$$\mathcal{L}_{\mathcal{SD}}(\mathbf{X},\mathbf{Y}) = \frac{1}{N}\sum_{i}\mathcal{L}_{\mathcal{CE}}(p(\mathbf{x}_i),\mathbf{y}_i) \tag{9}$$

$$\mathcal{L}_{\mathcal{ED}}(\mathbf{X_{ED}},\mathbf{Y}) = \frac{1}{N}\sum_{i}\mathcal{L}_{\mathcal{CE}}(p(\mathbf{x}_{\mathrm{ED}i}),\mathbf{y}_i) \tag{10}$$

$$\mathbf{X_{IS}} = w\mathbf{X} + (1-w)\mathbf{X_{ED}} \tag{11}$$

$$\mathcal{L}_{\mathcal{IS}}(\mathbf{X_{IS}},\mathbf{Y}) = \frac{1}{N}\sum_{i}\mathcal{L}_{\mathcal{CE}}(p(\mathbf{x}_{IS_i}),\mathbf{y}_i). \tag{12}$$

2) Regularization: In (13), supervised contrastive learning analogy to (7) is introduced to motivate D to learn class-wise domain invariant representation. Different from the above intra-class comparison, the inter-class comparison used here regards the same category and other different categories as positive and negative samples, respectively.

In addition, the intraclass comparison loss $\mathcal{L}_2$ is introduced again. Based on two purposes, one is to set the loss of optimizing G as a shared item to reduce the pressure of multitasking (D and G) optimization, and the other is to prompt intra-class and inter-class comparison learning to form a confrontation, for preventing the excessive aggregation or separation of SD and ED. The experimental results that do not use this confrontation are shown in the ablation experiment (no adv) of Table IV

$$\mathcal{L}_1 = -\sum_{i=0}^{N}\frac{1}{|P(i)|}\sum_{p\in P(i)}\log\frac{\exp\left(S\left(\mathbf{z}_i,\mathbf{z}_p^+\right)/\tau\right)}{\sum_{a\in N(i)}\exp\left(S\left(\mathbf{z}_i,\mathbf{z}_n^-\right)/\tau\right)}. \tag{13}$$

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C. Training Phase

G and D are optimized separately. First, D is optimized, including $\mathrm{FC}_{\mathrm{ext}}(\theta_f)$, $C(\theta_C)$ and $F(\theta_P)$ in (14), with SD and generated ED and IS as inputs. Where λ_1 and λ_2 are the hyper-parameters for balancing between interclass and intraclass supervision contrastive loss that constitutes the basic adversarial relationship

$$\min_{\theta_f,\theta_C,\theta_P} \mathcal{L}_{\mathcal{D}} = \mathcal{L}_{\mathcal{SD}} + \mathcal{L}_{\mathcal{ED}} + \mathcal{L}_{\mathcal{IS}} + \lambda_1\mathcal{L}_1 + \lambda_2\mathcal{L}_2. \quad (14)$$

To optimize $G(\theta_G)$, (10) is used as an auxiliary loss to ensure ED in the correct label space in (15), where λ_2 is a hyper-parameter for balancing the intra-class comparison loss and cross entropy

$$\min_{\theta_G} \mathcal{L}_{\mathcal{G}} = \mathcal{L}_{\mathcal{ED}} + \lambda_2\mathcal{L}_2. \quad (15)$$

The G, in particular, generates "fake" samples from SD on a continuous basis, but while the D distinguishes SD and ED via $\mathcal{L}_2$ and enhances its own classification ability. Similar to generative adversarial networks (GANs) [36], the two "develop" together to build the second confrontation relationship, eventually reaching a delicate equilibrium, constraining the nonredundant and reliable expansion of SD while ensuring the sensitivity of the D to class separable features.

D. Generalization Performance

Fig. 4 shows the visualization results of the features extracted by discriminator D on SD, ED, and TD following t-SNE [37] using PaviaU as SD and PaviaC as TD. When compared to Fig. 4(a) and (b), SD samples produced using $\mathrm{FC}_{\mathrm{ext}}$ perform much better at inter-class separability under low-dimensional features. As illustrated in Fig. 4(b) and (d), ED guarantees not only the nonredundant effective growth of SD but also class separability. According to Fig. 4(e) and (f), for the unseen TD, the low-dimensional features of distinct categories are highly separable, indicating that the discriminator's generalization capacity is solid under the effect of sample augmentation and regularization.

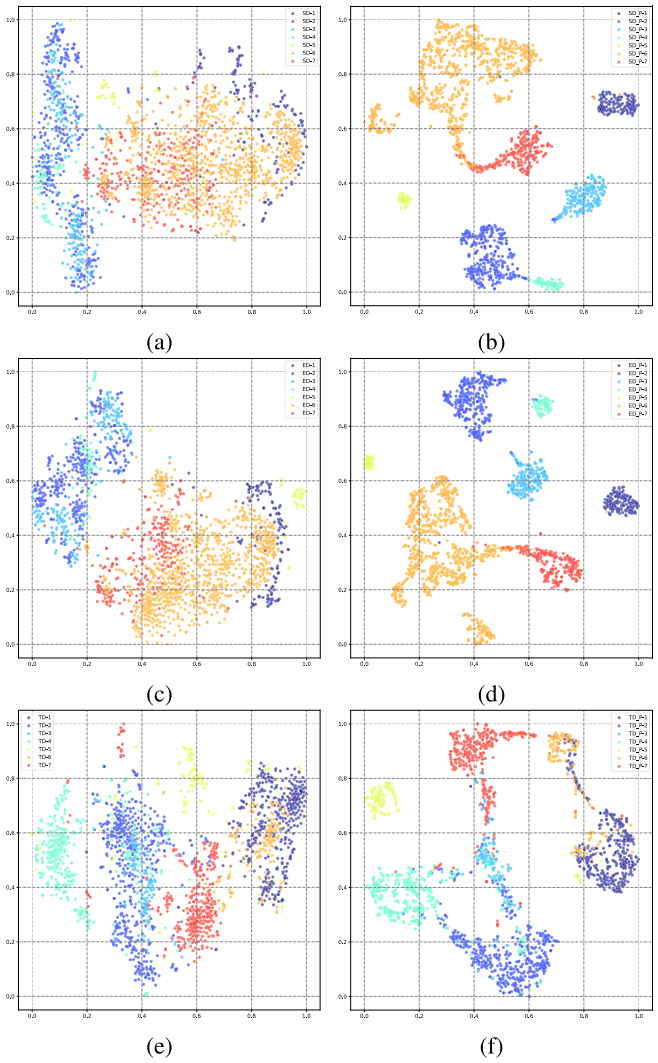


Fig. 4. T-SNE of class separability of the LLURnet using the Pavia dataset, where SD is the PaviaU data, TD is the PaviaC data, and the number represents class index. (a) Raw sample of SD. (b) SD features by P. (c) Samples of ED by G. (d) ED features by P. (e) Raw samples of TD. (f) TD features by P.

IV. Experimental Results and Discussion

Experiments are based on three cross-scene HSI datasets, Houston, Pavia, and cross-city dataset Shanghai-Hangzhou. The SD in the Houston dataset is strengthened four times by random flip and random radiation noise (illumination), whereas the others are not. Part of the representative DG algorithm in the field of CV, the state-of-the-art (SOTA) DA and DG methods for HSIC are used for comparative experiments. To assess classification performance, the overall accuracy (OA), average accuracy (AA), and the Kappa coefficient (KC) are calculated. The testing results are shown as mean $\pm$ standard deviation (STD) by repeated experiments five times in order to evaluate the uncertainty of the proposed method, which is primarily due to the randomization process of the generator during the training process, the random division of the datasets, and the random initialization of the network cannot be ignored. PyTorch and an Nvidia GTX 3060 GPU with 12 GB is the basic software and hardware platform.

A. Experimental Data

1) Houston Dataset: The dataset includes scenes from Houston in Debes et al. [38] and Houston in Le Saux et al. [39], which were collected by various sensors on the campus and neighborhood of the University of Houston in various years. The Houston 2013 dataset has an image resolution of 2.5 m, 349 $\times$ 1905 pixels, 144 spectral bands, and a wavelength range of 380–1050 nm. The Houston 2018 dataset comprises 48 spectral bands and the same wavelength range, and the image has a 1 m spatial resolution. In their scene, there are seven constant classes. We choose the overlapping area of 209 $\times$ 955 and extract 48 spectral bands (wavelength range 0.38–1.05 μm) from the Houston 2013 scene that correspond to the Houston 2018 scene. The classes and the number of samples are shown in Table I. The false-color and ground truth maps are shown in Fig. 5.

2) Pavia Dataset: The Pavia Center (PC) and the University of Pavia (UP) are contained in the Pavia dataset. reflective optics spectrographic image system (ROSIS), which has

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TABLE I

NUMBER OF SOURCE AND TARGET SAMPLES FOR THE HOUSTON DATASET

Class		Number of Samples	
No.	Name	Houston 2013 (Source)	Houston 2018 (Target)
1	Grass healthy	345	1353
2	Grass stressed	365	4888
3	Trees	365	2766
4	Water	285	22
5	Residential buildings	319	5347
6	Non-residential buildings	408	32459
7	Road	443	6365
Total		2530	53200

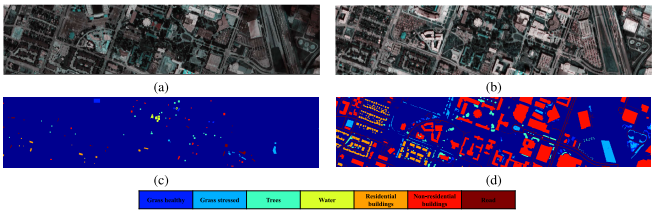


Fig. 5. Pseudo-color image and ground truth map [16], [40] of Houston dataset. (a) Pseudo-color image of Houston 2013. (b) Pseudo-color image of Houston 2018. (c) Ground truth map of Houston 2013. (d) Ground truth map of Houston 2018.

TABLE II

NUMBER OF SOURCE AND TARGET SAMPLES FOR THE PAVIA DATASET

Class		Number of Samples	
No.	Name	UP (Source)	PC (Target)
1	Tree	3064	7598
2	Asphalt	6631	9248
3	Brick	3682	2685
4	Bitumen	1330	7287
5	Shadow	947	2863
6	Meadow	18649	3090
7	Bare soil	5029	6584
Total		39332	39355

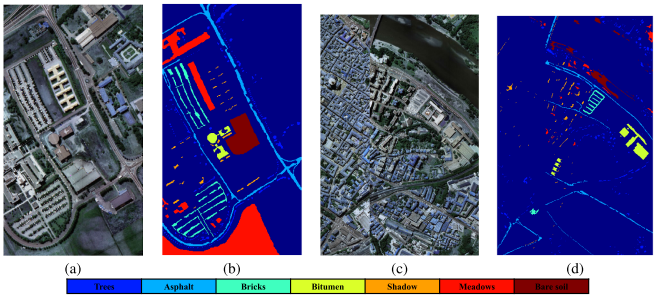


Fig. 6. Pseudo-color image and ground truth map [16], [40] of Pavia dataset. (a) Pseudo-color image of UP. (b) Ground truth map of UP. (c) Pseudo-color image of PC. (d) Ground truth map of PC.

spectral coverage from 430 to 860 nm, was used to shoot. The PC has 102 bands and 1096 × 715 pixels. The UP includes 103 spectral bands, 610 × 340 pixels, and a spatial resolution of 1.3 m. The last band has been deleted to maintain parity with the PC in terms of spectral band. Their labels are the same and provided in Table II. Their ground-truth maps and false-color photos are shown in Fig. 6.

3) Shanghai-Hangzhou Dataset: The Shanghai-Hangzhou datasets were captured by the EO-1 Hyperion hyperspectral sensor, which retains 198 bands. The size of the Shanghai

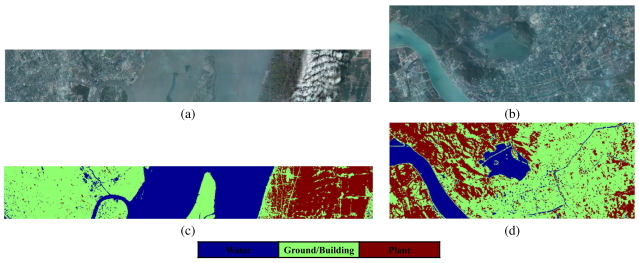


Fig. 7. Pseudocolor image and ground-truth map of S-H [16]. (a) Pseudocolor image of Shanghai. (b) Pseudocolor image of Hangzhou. (c) Ground-truth map of Shanghai. (d) Ground-truth map of Hangzhou.

TABLE III

NUMBER OF SOURCE AND TARGET SAMPLES FOR THE SHANGHAI-HANGZHOU DATASET

Class		Number of Samples	
No.	Name	Hangzhou (Source)	Shanghai (Target)
1	Water	18043	123123
2	Land/Building	77450	161689
3	Plant	40207	83188
Total		135700	368000

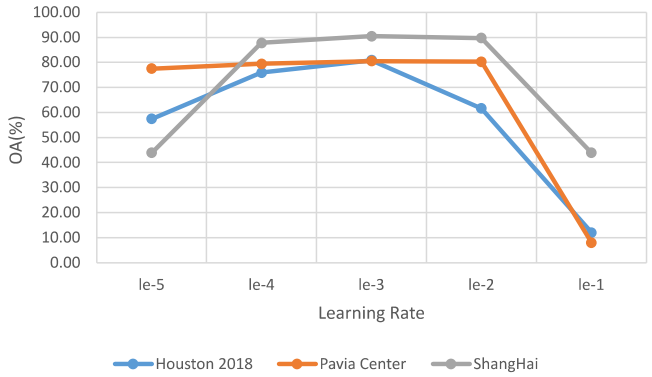


Fig. 8. Parameter tuning of the base learning rate for the proposed LLURnet using the three TDs.

data image is 1600 × 230. The Hangzhou data image size is 590 × 230. The straightline distance between Shanghai and Hangzhou is 162 km. Table III lists three common land cover classes for cross-scene categorization, and their pseudocolor representations and ground-truth maps are illustrated in Fig. 7.

B. Parameter Sensitivity Analysis

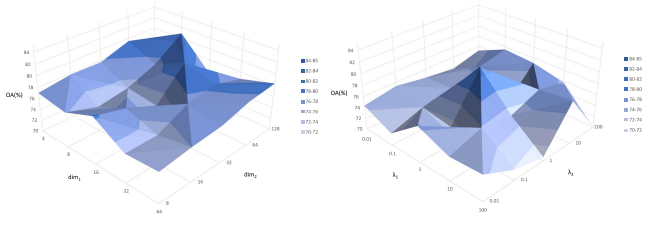
To assess the sensitivity of LLURnet on the two TDs, a parameter sensitivity analysis is performed. The base learning rate in Fig. 8, regularization parameters λ_1/λ_2 and embedding feature dimension $\dim_1/\dim_2$, regarded as adjustable hyperparameters are selected from $\{1e-5, 1e-4, 1e-3, 1e-2, 1e-1\}$, $\{1e-2, 1e-1, 1e+0, 1e+1, 1e+2\}$, and $\{4, 8, 16, 32, 64, 128\}$, respectively.

According to Figs. 9–11, for the same TD, the STD of OA under varied λ_1/λ_2 is larger than $\dim_1/\dim_2$ [2.35 in Fig. 9(b) > 1.65 in Fig. 9(a), 2.26 in Fig. 10(b) > 2.05 in Fig. 10(a), 2.61 in Fig. 11(b) > 1.95 in Fig. 11(a)], indicating that within a given value range, the model's generalization

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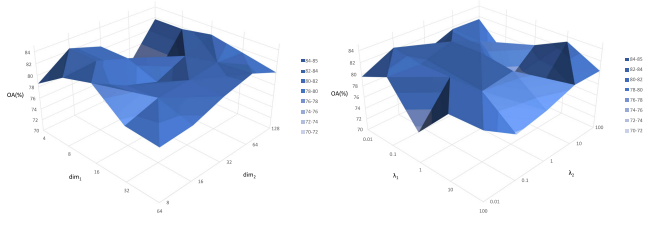
TABLE IV
ABLATION COMPARISON OF EACH VARIANT OF LLURNET

Model	FC_{ext}	LLURnet (no linear)	LLURnet (no spa & spe)	LLURnet (no random)	LLURnet (no $\mathcal{L}_1$)	LLURnet (no $\mathcal{L}_2$)	LLURnet (no adv)	LLURnet
Dataset	Houston 2018(Target)							
OA(%)	73.5 ± 1.19	67.7 ± 3.82	72.33 ± 0.79	75.27 ± 0.22	70.78 ± 0.85	75.81 ± 1.57	71.8 ± 2.63	**80.88 ± 1.88**
AA(%)	69.26 ± 0.74	53.54 ± 4.09	72.21 ± 1.31	76.06 ± 0.37	67.16 ± 0.04	69.73 ± 1.25	74.95 ± 1.41	**78.59 ± 2.95**
KC(κ)	57.94 ± 1.20	37.89 ± 7.35	57.93 ± 0.67	61.08 ± 0.60	55.41 ± 0.94	60.16 ± 2.40	57.44 ± 2.61	**68.43 ± 2.46**
Dataset	Pavia Center(Target)							
OA(%)	81.76 ± 1.26	77.67 ± 1.98	78.5 ± 1.61	79.85 ± 1.28	80.02 ± 2.63	78.84 ± 1.22	75.88 ± 2.78	**83.53 ± 1.17**
AA(%)	80.01 ± 1.37	77.13 ± 2.15	75.72 ± 1.87	78.36 ± 0.89	79.5 ± 2.29	76.7 ± 1.91	75.39 ± 2.02	**81.75 ± 1.19**
KC(κ)	78.07 ± 1.50	73.3 ± 2.36	74.17 ± 1.92	75.78 ± 1.49	76.1 ± 3.12	74.5 ± 2.51	71.06 ± 3.32	**80.17 ± 1.26**
Dataset	ShangHai(Target)							
OA(%)	84.06 ± 5.38	87.48 ± 2.37	85.51 ± 5.20	87.62 ± 1.84	86.35 ± 1.55	85.69 ± 3.30	87.02 ± 1.32	**89.46 ± 0.84**
AA(%)	87.42 ± 4.05	89.19 ± 1.46	88.51 ± 3.95	89.99 ± 1.46	88.98 ± 1.18	88.56 ± 2.56	89.64 ± 1.04	**91.28 ± 0.66**
KC(κ)	76.39 ± 7.46	81.05 ± 3.36	78.44 ± 7.27	81.33 ± 2.68	79.47 ± 2.22	78.59 ± 4.74	80.49 ± 1.91	**83.97 ± 1.24**



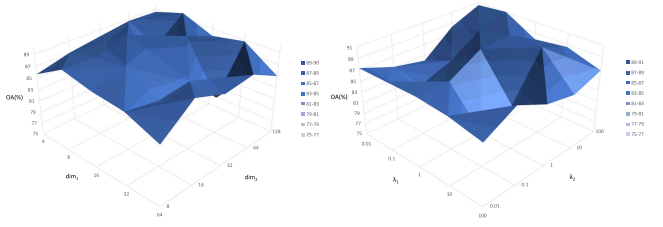
(a) (b)

Fig. 9. Parameter tuning of λ and dim for the LLURnet om TD = Houston18. (a) Tuning of $\text{dim}_1/\text{dim}_2$, STD = 1.65. (b) Tuning of λ_1/λ_2, STD = 2.35.



(a) (b)

Fig. 10. Parameter tuning of λ and dim for the LLURnet on TD = PaviaC. (a) Tuning of $\text{dim}_1/\text{dim}_2$, STD = 2.05. (b) Tuning of λ_1/λ_2, STD = 2.26.



(a) (b)

Fig. 11. Parameter tuning of λ and dim for the LLURnet on TD = Shanghai. (a) Tuning of $\text{dim}_1/\text{dim}_2$, STD = 1.95. (b) Tuning of λ_1/λ_2, STD = 2.61.

stability is more sensitive to the regularization parameter than to the embedded feature dimension. The cause could be that LLURnet is an over-parameterized network. Despite the fact that the network's loss landscape is significantly nonconvex, it exhibits great geometric characteristics in the small neighborhood near the initialization point. That is, near the initialization point, even though the network output is extremely nonlinear to the input data, the network output has an almost linear relationship to the network parameters. As a result, when doing nonconvex optimization, the neighborhood near the initialization point is actually performing convex optimization. Thus, the weight of the regularization item (λ_1/λ_2) is more important than the embedded feature dimension ($\text{dim}_1/\text{dim}_2$) which mainly impacts the parameter quantity.

C. Ablation Study

Linear resampling, 3D-CNN-based spatial-spectral style extraction (spa & spe), and randomization are crucial components of the generator, and the major approaches for optimizing discriminator and generator include interclass/intraclass comparison learning and adversarial training. Ablation analyses are carried out to investigate the contribution of LLURnet core elements by removing each component from the entire framework in Table IV.

There are seven variants in the ablation analyses: 1) FC_{ext}: the model of testing stage in Fig. 2 is training on SD and directly applied to TD; 2) "LLURnet (no linear)": the linear layer(Conv2D & DeCon2D) is deleted from generator; 3) "LLURnet (no spa & spe)": the Conv3D&DeConv3D is deleted; 4) "LLURnet (no random)": the style random is removed; 5) "LLURnet (no $\mathcal{L}_1$)": the $\mathcal{L}_1$ in (14) is removed; 6) "LLURnet (no $\mathcal{L}_2$)": the $\mathcal{L}_2$ in (14) and (15) is removed; and 7) "LLURnet (no adv)": the $\mathcal{L}_2$ in (14) is removed but reserved in (15).

The ablation experiment demonstrates the importance of each component, but the generalization contribution of specific module on the Houston and Pavia datasets differs.

For TD = Houston18, the OA after ablation is basically maintained at 73% ± 2% except for "no linear," indicating that the samples in Houston pay more attention to linear resampling (including down sampling and up sampling) to feature scaling of the spectral dimension. One of the underlying issues could be that the training data from Houston13 only has 48 bands (Houston13 has 144 spectral bands originally. The truncation operation was used to generate the training set in order to align the 48 bands of Houston18), so the average information of each band is much larger than Pavia's 102 bands.

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TABLE V

CLASS-SPECIFIC AND AVERAGE/OVERALL CLASSIFICATION ACCURACY (%) OF DIFFERENT METHODS FOR THE TARGET SCENE HOUSTON 2018 DATA

Class	Classification algorithms									
	DAAN [42]	MRAN [43]	DSAN [44]	HTCNN [45]	PDEN [46]	LDSDG [19]	SagNet [21]	SDEnet [17]	LDGnet [18]	LLURnet
1	68.29	41.02	62.31	11.83	46.49	10.13	25.79	24.69	61.71	78.79
2	77.80	76.94	77.50	70.11	77.60	62.97	62.79	84.98	77.45	87.48
3	67.50	65.91	74.55	54.99	59.73	60.81	48.66	59.65	62.08	58.79
4	100	100	100	54.55	100	81.82	81.82	100	95.45	100
5	47.69	36.90	73.39	55.60	49.62	45.65	59.57	62.33	69.53	76.36
6	79.49	82.68	86.84	92.85	84.98	89.22	89.28	90.54	91.57	86.20
7	45.12	56.43	46.33	46.47	64.21	44.15	34.99	57.45	45.42	62.48
AA(%)	69.41	65.70	74.42	55.20	68.95	56.39	57.56	68.52	71.89	**78.59**
OA(%)	70.45 ± 1.54	72.11 ± 1.79	78.21 ± 1.05	77.67 ± 2.11	75.40 ± 1.76	74.23 ± 1.49	73.01 ± 1.81	79.96 ± 1.18	80.34 ± 1.74	**80.88 ± 1.88**
KC(κ)	53.86 ± 2.14	55.03 ± 2.86	64.10 ± 1.83	60.18 ± 3.07	55.87 ± 2.92	55.42 ± 2.27	55.29 ± 3.50	65.15 ± 2.24	65.80 ± 2.59	**68.43 ± 2.46**

TABLE VI

CLASS-SPECIFIC AND AVERAGE/OVERALL CLASSIFICATION ACCURACY (%) OF DIFFERENT METHODS FOR THE TARGET SCENE PC DATA

Class	Classification algorithms									
	DAAN [42]	MRAN [43]	DSAN [44]	HTCNN [45]	PDEN [46]	LDSDG [19]	SagNet [21]	SDEnet [17]	LDGnet [18]	LLURnet
1	71.98	59.16	93.93	96.06	85.93	91.09	98.35	89.93	95.20	86.44
2	78.98	85.15	79.8	57.70	88.56	73.51	59.76	81.22	82.79	87.1
3	19.37	46.18	53.97	2.76	61.34	2.23	5.40	72.77	80.48	77.5
4	58.67	69.58	75.75	93.25	85.49	71.72	87.03	82.54	85.11	84.82
5	70.87	64.58	99.44	89.94	87.95	71.04	93.19	84.81	93.15	85.02
6	83.07	89.22	74.43	70.97	79.26	57.12	49.81	75.11	66.93	69.09
7	55.59	60.10	67.31	42.28	64.75	78.13	57.94	78.74	81.97	82.31
AA(%)	62.65	67.71	77.80	64.71	79.04	63.55	64.50	80.73	83.66	**81.75**
OA(%)	66.71 ± 2.15	69.01 ± 1.21	78.21 ± 1.10	68.31 ± 1.62	80.27 ± 1.38	70.88 ± 1.55	69.28 ± 1.11	81.94 ± 1.55	82.53 ± 1.24	**83.53 ± 1.17**
KC(κ)	59.76 ± 3.47	63.04 ± 1.90	74.10 ± 1.57	62.17 ± 2.57	76.81 ± 1.92	64.06 ± 2.07	63.04 ± 1.67	78.33 ± 2.47	78.85 ± 1.78	**80.17 ± 1.26**

For TD = PaviaC, the OA after ablation is basically maintained at 80% ± 2% except "no adv," indicating that Pavia requires moderate intra-class/inter-class confrontation to balance the distance between SD and ED in the feature space, for reducing the risk of overfitting while reserve certain domain-specific information, one of the potential causes may be the number of training samples, which is 31 465 in PaviaU but only 2024 in Houston13, while they both have the same 7 categories.

Additionally, consider the generalization gap between the FC_{ext} and the entire LLURnet. LLURnet uses the generator and double confrontation training to increase the OA by 7.38% for TD = Houston13, but only 1.77% for TD = PaviaC. This is due to the difference between the SD and TD of the two datasets strictly. Analyzed from the perspective of SD, as shown in Figs. 5(c) and 6(b), Houston13 labeled samples, in terms of number and quality, it clearly lags below PaviaU. The former labeled ones are few, fragmentary, and scattered, whereas the latter are numerous and well integrated. For SD = Houston13, LLURnet's data augmentation and regularization approach simply "rescued" its deficit. Instead, PaviaU's category feature expression is more explicit, so LLURnet's "interpolation" (generation) of the sample space is easily superfluous.

Finally, for TD = Shanghai, it is worth noting that its SD comprises a large number of training samples (135 700), and only three almost unrelated categories need to be distinguished. From a decision-making standpoint, this task is "easier" than TD = Houston18/PaviaC. As shown in Table IV, only the basic FC_{ext} network achieves 84.06% OA, while the STD is as high as 5.38%. In comparison, the OA of LLURnet improves by 5.4%, simultaneously, the STD is reduced by 4.54%, demonstrating that the proposed generator and regularization strategy not only enhances model training robustness, but also enables the discriminator FC_{ext} to obtain more reliable domain-invariant features.

D. Performance on Cross-Scene HSI Classification

To compare the performance of LLURnet, relevant algorithms such as DA methods including DAAN [41], MRAN [42], DSAN [43], and HTCNN [44], DG methods including PDEN [45], LDSDG [19], SagNet [21], SDEnet [17], and LDGnet [18] are used. The patch size of LDSDG and SagNet is set to 32 × 32 to match the Resnet18 input size, while the patch size of other networks for training and testing is unified to 13 × 13. The optimizer uses Adaptive Moment Estimation (Adam). The data of SD = Houston13/PaviaU with labels are set 80% for training and 20% for validation. The data of SD = Hangzhou is set 5% for training and 95% for validation. Since the preprocessing strategy and test conditions are consistent, for SD = Houston13/PaviaU, the results on DAAN, MRAN, DSAN, HTCNN, PDEN, LDSDG, SagNet, and SDEnet are from [17], and the results on LDGnet are from [18]). For SD = Hangzhou, the results on DAAN, MRAN, DSAN, and HTCNN are from [16]. The optimal base learning rate, regularization parameters, and embedding feature dimension of LLURnet are selected from $\{1e-5, 1e-4, 1e-3, 1e-2, 1e-1\}$, $\{1e-2, 1e-1, 1e+0, 1e+1, 1e+2\}$, and $\{4, 8, 16, 32, 64, 128\}$, respectively. To identify the corresponding optimal parameters, cross-validation is performed.

According to Tables V–VII, The OA of the LLURnet is 2%–5% higher or equivalent than the best DA algorithm

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TABLE VII
CLASS-SPECIFIC AND AVERAGE/OVERALL CLASSIFICATION ACCURACY (%) OF DIFFERENT METHODS FOR THE TARGET SCENE SHANGHAI DATA

Class	Classification algorithms									
	DAAN [42]	MRAN [43]	DSAN [44]	HTCNN [45]	PDEN [46]	LDSDG [19]	SagNet [21]	SDEnet [17]	LDGnet [18]	LLURnet
1	92.11	85.36	91.90	89.24	93.58 ± 1.34	89.26 ± 4.03	93.01 ± 1.69	93.89 ± 1.04	92.7 ± 2.78	92.11 ± 1.61
2	77.73	82.06	84.83	37.14	77.00 ± 6.03	72.33 ± 7.08	73.10 ± 4.24	73.75 ± 7.52	75.92 ± 6.93	82.34 ± 2.12
3	99.32	99.62	95.45	100	99.49 ± 0.22	99.40 ± 0.65	99.57 ± 0.19	99.57 ± 0.19	99.43 ± 0.23	99.40 ± 0.20
AA(%)	89.72	89.01	90.72	75.46	90.02 ± 2.17	87.00 ± 1.08	88.56 ± 1.48	89.07 ± 2.36	89.35 ± 2.56	**91.28 ± 0.66**
OA(%)	87.42	87.13	89.6	68.78	86.59 ± 2.95	84.51 ± 1.99	85.75 ± 1.92	86.33 ± 3.17	86.85 ± 3.28	**89.46 ± 0.84**
KC(κ)	80.99	80.39	84.08	55.34	81.23 ± 4.25	76.75 ± 2.78	78.62 ± 2.75	79.52 ± 4.47	80.22 ± 4.73	**83.97 ± 1.24**

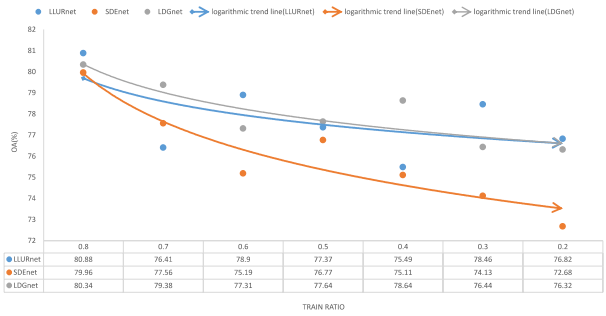


Fig. 12. Comparison of OA on TD = Houston18 under different percentages of training samples.

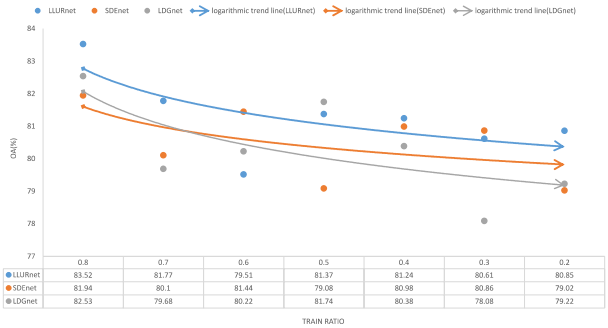


Fig. 13. Comparison of OA on TD = PaviaC under different percentages of training samples.

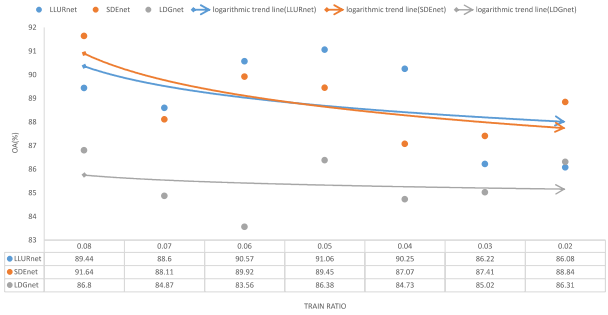


Fig. 14. Comparison of OA on TD = Shanghai under different percentages of training samples.

DSAN on three TDs, indicating that, while the DA strategy obtained more prior information on TD, the DA models for domain alignment are not as accurate as the latent variable learning based on HSI features. On the one hand, conventional DA algorithms may not make adequate use of TD data. Yet, these DA algorithms may place too much emphasis on the unsupervised aspects of TD, while the more reliable supervised learning in SD is underutilized.

When compared to the DG methods in CV, including PDEN, LDSDG, and SagNet. the OA of LLURnet is increased by 3%–14%. This enormous gap is mostly due to discrepancies in feature extraction architecture (encoders, decoders, etc.). These methods created for RGB images rely on pixel spatial distribution, however, HSI spatial information has a more limited contribution to classification and the spectrum contains crucial elements of the category, which is a primary cause of methodology direct transfer inapplicability.

Compared with SDEnet, under TD = Houston18 in Fig. 12, with the reduction of training samples (from 2024 to 506), the OA of LLURnet declined by 4.06%, while the OA of SDEnet reduced by 7.28%. With the reduction of training samples in TD = PaviaC in Fig. 13 (from 31 465 to 7866), the OA of LLURnet declined by 2.67% and the OA of SDEnet dropped by 2.92%. With the reduction of training samples in TD = Shanghai in Fig. 14 (from 10 856 to 2714), the OA of LLURnet declined by 3.36% and the OA of SDEnet dropped by 2.8%. Referring to the model structure and optimization approach, the key distinction is in the generation of latent variables in the generator and the differences in confrontation training scenarios. To some extent, the experimental results demonstrate that the LLURnet generator and dual confrontation training are superior.

Compared with LDGnet, under TD = Houston18 in Fig. 12, the declining trend of LLURnet is nearly identical to LDGnet. LLURnet's logarithmic trend line is higher than LDGnet's under TD = PaviaC in Fig. 13. For TD = Shanghai in Fig. 14, because the dataset has only three labels with nearly no semantic association, the OA of LDGnet is always lower, however, with the pretraining parameters in LDGnet, its OA hardly falls with the decrease of the percentage of training samples.

In Figs. 15–17, classification maps are used to display the prediction results. In contrast, the spatial distribution of LLURnet prediction results is consistent (no noticeable noise), and the category borders are unambiguous. It demonstrates the network's ability to learn spatial associations and sensitivity to extracting class-separable features.

Table VIII shows the total trainable parameter size and testing execution time with the same batch size 256 to demonstrate the computational complexity of various techniques.

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TABLE VIII
COMPARISON OF TOTAL TRAINABLE PARAMETER SIZE AND TESTING EXECUTION TIME

Methods	DAAN [42]	MRAN [43]	DSAN [44]	HTCNN [45]	PDEN [46]	LDSDG [19]	SagNet [21]	SDEnet [17]	LDGnet [18]	LLURnet
Dataset	Houston 2018(Target)									
Total Trainable Params (MB)	33.09	25.18	25.01	2.81	4.65	11.39	20.81	1.79	34.22	**0.55**
Testing Time(s)	11.36	16.13	10.27	9.43	2.18	7.31	6.33	1.69	16.96	**1.68**
Dataset	Pavia Center(Target)									
Total Trainable Params (MB)	33.25	25.35	25.17	2.81	4.81	11.65	20.97	2.41	35.10	**0.57**
Testing Time(s)	13.31	17.93	12.33	12.16	2.43	9.52	8.49	2.61	14.23	**2.54**
Dataset	ShangHai(Target)									
Total Trainable Params (MB)	28.52	25.63	25.46	2.81	5.09	12.12	21.25	3.50	336.60	**0.63**
Testing Time(s)	213.70	294.06	202.52	206.56	50.27	145.38	145.68	53.45	256.96	**53.90**

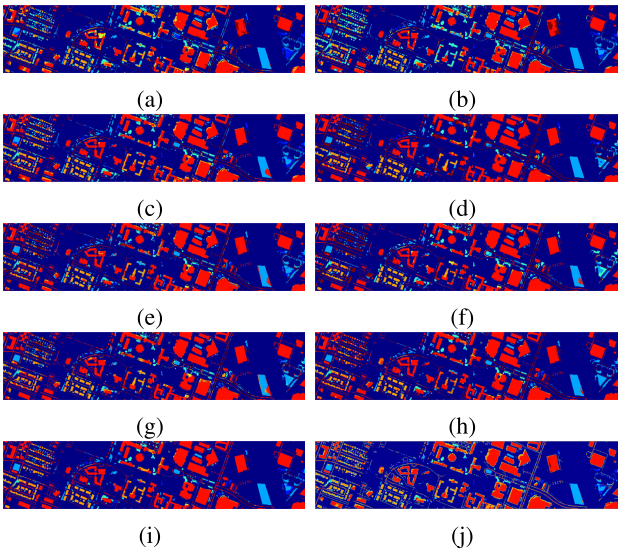


Fig. 15. Data visualization and classification maps for target scene Houston 2018 data obtained with different methods including: (a) DAAN (71.13%), (b) MRAN (72.48%), (c) DSAN (78.52%), (d) HTCNN (77.42%), (e) PDEN (75.98%), (f) LDSDG (73.55%), (g) SagNet (73.64%), (h) SDEnet (80.11%), (i) LDGnet(80.55%), and (j) LLURnet(80.92%).

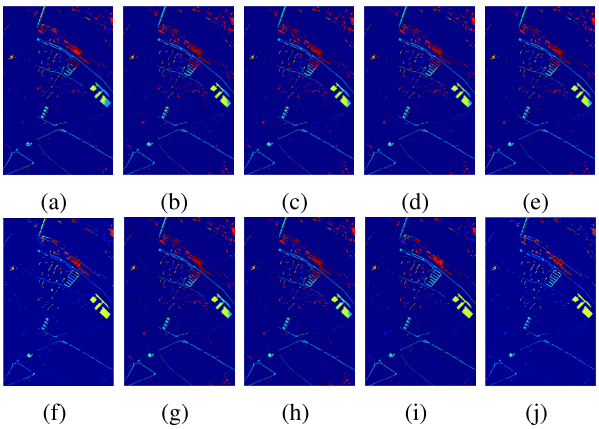


Fig. 16. Data visualization and classification maps for target scene PC data obtained with different methods including: (a) DAAN (65.62%), (b) MRAN (69.22%), (c) DSAN (78.94%), (d) HTCNN (68.75%), (e) PDEN (80.87%), (f) LDSDG (71.02%), (g) SagNet (69.90%), (h) SDEnet (81.76%), (i) LDGnet(82.84%), and (j) LLURnet(83.78%).

It can be shown that the LLURnet benefits from a structurally streamlined processing hierarchy, which has quite lower time and space computational cost than others using VGG or Resnet as the backbone.

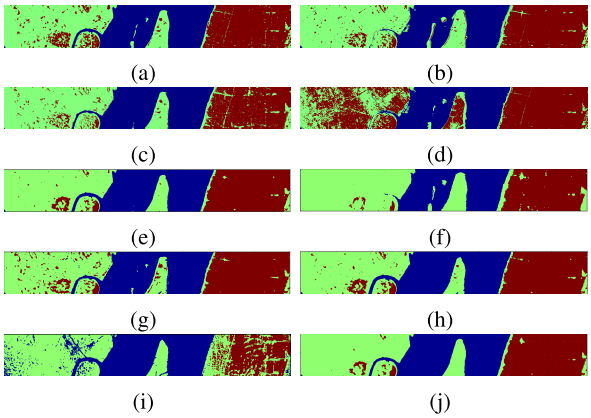


Fig. 17. Data visualization and classification maps for target scene Shanghai data obtained with different methods including: (a) DAAN (87.42%), (b) MRAN (87.13%), (c) DSAN (89.60%), (d) HTCNN (68.78%), (e) PDEN (84.12%), (f) LDSDG (87.43%), (g) SagNet (82.72%), (h) SDEnet (88.46%), (i) LDGnet(89.92%), and (j) LLURnet(90.50%).

V. CONCLUSION

In order to generalize to the TD only by training on the SD, a locally linear unbiased randomization network for cross-scene HSIC is proposed. Specifically, the generator with linear resampling and style randomization is designed, for generation efficacy and reliability. Besides, a double-adversarial regularization training for generator and discriminator is suggested, which accomplishes nonredundant expansion and alleviates the contradiction between aggregation and separation with lower computational cost. The LLURnet outperforms the non-HSI specialized DG algorithms and the representative HSI DA approaches. Robustness to the number of training samples is on par with or better than the present HSIC DG networks. In general, as with the cross-domain HSIC, the design and optimization principles of LLURnet demonstrate the practicality and promise of the framework focus on data augmentation and feature space regularization.

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